Reg. No.: _

Amrita Vishwa Vidyapeetham, Kochi Campus
School of Computing
End Semester Examination, November / December 2023
2022 BCA / Integrated MCA – III Semester

21CSA202 Object Oriented Programming Using Java

Answer all Questions

Time: 3 Hours **Maximum Marks: 100**

PART A

(10 x 2 marks = 20 marks)

1. What are access modifiers in java? Explain the different types of access modifiers?
2. Identify the error in the following code and state the reason for the same:

```
public class Test
{
  Test() {
     m2();
  }
 void m1() {
      System.out.println("Instance method");
  }
 static void m2() {
      System.out.println("Static method");
      m1();
  }
public static void main(String[] args)
{
    new Test();
 }
}
```

3. Describe the usage of final keyword in java.
4. Write a Java program to accept two numbers using command line arguments and show the result after performing all the arithmetical calculations. It should also handle the division by zero exception.
5. Find the output of the below program?

```
public class Test extends Thread {
  public static void main(String[] args) {
    System.out.print(Thread.currentThread().getPriority()+" ");
    Thread.currentThread().setPriority(9);
    Test t = new Test();
    System.out.println(t.getPriority());
}
}
```

6. Identify the error and debug the following code.

```
class Multi3 implements Runnable{
public void run(){
System.out.println("thread is running...");
```

```
}

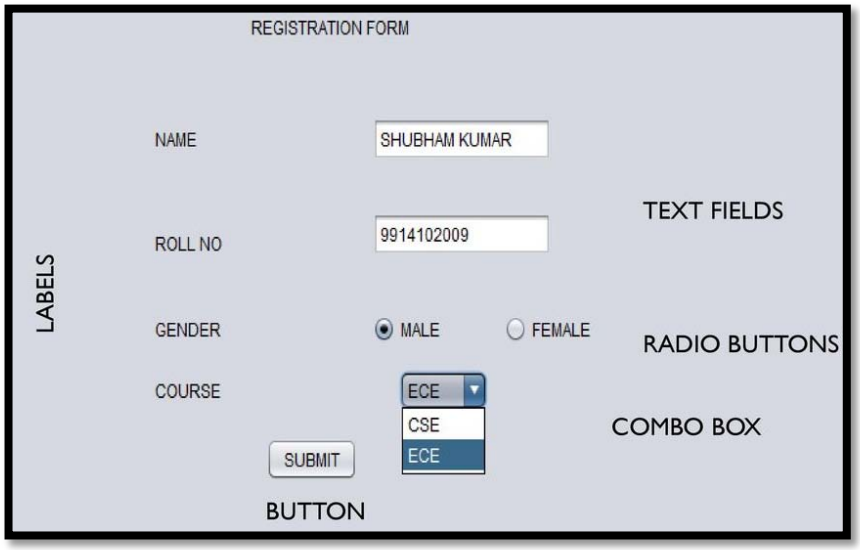
class MultiDemo{
public static void main(String args[]){
Thread t1 =new Thread();
t1.start();
 }
}
```

7. What is the role of layout manager in java? What are the different types of layout managers in Java?
8. List the differences between AWT and Swing.
9. Define UML Diagram? What are the main types of UML diagrams?
10. Draw a simple Use case diagram for bank ATM system

PART B

(10 x 8 marks = 80 marks)

11. Elucidate the important features of java?
12. Write a program to print the name, salary and date of joining of 10 employees in a company. [use array object]
13. Write a Java program to create a user defined package in Java.
14. Write a java program to demonstrate dynamic polymorphism.
15. Describe the various types of inheritance in java. Write down a program which depicts the multilevel inheritance.
16. Elucidate the Thread life cycle in java with a suitable diagram.
17. Write a Java program to create and write the string content into a file. The program should read and display the same and throws an exception if the file is not found.
18. Write a java program to create the following user interface using swing / awt in java.



19. Write short notes on collection interface in java? Write a java program to add elements to an ArrayList and retrieve and print the same.
20. Draw an UML Activity diagram for a library management system.

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